# EDA USING SQL

## EDA

EDA is the process of looking at and understanding your data before doing any serious analysis, so you know what's going on inside it.

### Schema of Tables

```sql
-- Check the structure of Scheme of the data
SELECT * FROM INFORMATION_SCHEMA.TABLES
```

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Results Messages

| | TABLE_CATALOG | TABLE_SCHEMA | TABLE_NAME | TABLE_TYPE |
|---|---|---|---|---|
| 1 | DataWareHouseAnalytics | dbo | dim_customers | BASE TABLE |
| 2 | DataWareHouseAnalytics | dbo | dim_products | BASE TABLE |
| 3 | DataWareHouseAnalytics | dbo | fact_sales | BASE TABLE |

### Schema of Columns of Each table

```sql
--Explore all the columns in the database
SELECT * FROM INFORMATION_SCHEMA.COLUMNS

--ALL COLUMNS FROM A SPECIFIC TABLE
SELECT * FROM INFORMATION_SCHEMA.COLUMNS
WHERE TABLE_NAME='dim_customers'
```

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Results Messages

| | TABLE_CATALOG | TABLE_SCHEMA | TABLE_NAME | COLUMN_NAME | ORDINAL_POSITION | COLUMN_DEFAULT | IS_NULLABLE | DATA_TYPE | CHARACTER_MAXIMUM_LENGTH |
|---|---|---|---|---|---|---|---|---|---|
| 1 | DataWareHouseAnalytics | dbo | dim_customers | customer_key | 1 | NULL | YES | smallint | NULL |
| 2 | DataWareHouseAnalytics | dbo | dim_customers | customer_id | 2 | NULL | YES | smallint | NULL |
| 3 | DataWareHouseAnalytics | dbo | dim_customers | customer_number | 3 | NULL | YES | nvarchar | 50 |
| 4 | DataWareHouseAnalytics | dbo | dim_customers | first_name | 4 | NULL | YES | nvarchar | 50 |
| 5 | DataWareHouseAnalytics | dbo | dim_customers | last_name | 5 | NULL | YES | nvarchar | 50 |
| 6 | DataWareHouseAnalytics | dbo | dim_customers | country | 6 | NULL | YES | nvarchar | 50 |
| 7 | DataWareHouseAnalytics | dbo | dim_customers | marital_status | 7 | NULL | YES | nvarchar | 50 |
| 8 | DataWareHouseAnalytics | dbo | dim_customers | gender | 8 | NULL | YES | nvarchar | 50 |
| 9 | DataWareHouseAnalytics | dbo | dim_customers | birthdate | 9 | NULL | YES | date | NULL |
| 10 | DataWareHouseAnalytics | dbo | dim_customers | create_date | 10 | NULL | YES | date | NULL |

### Country from where Customers comes From

```sql
--Check all country from where the customers comes from
SELECT DISTINCT
country
FROM dim_customers
```

100 % No issues found

Results Messages

| | country |
|---|---|
| 1 | n/a |
| 2 | Germany |
| 3 | United States |
| 4 | Australia |
| 5 | United Kingdom |
| 6 | Canada |
| 7 | France |

### Category & subcategory Divison

```sql
--Explore All Product Categories "The Major Divisions"
SELECT DISTINCT category,subcategory,product_name FROM dim_products
ORDER BY 1,2,3

--DATE EXPLORATION
```

100 % No issues found

Results Messages

| | category | subcategory | product_name |
|---|---|---|---|
| 1 | NULL | NULL | HL Mountain Pedal |
| 2 | NULL | NULL | HL Road Pedal |
| 3 | NULL | NULL | LL Mountain Pedal |
| 4 | NULL | NULL | LL Road Pedal |
| 5 | NULL | NULL | ML Mountain Pedal |
| 6 | NULL | NULL | ML Road Pedal |
| 7 | NULL | NULL | Touring Pedal |
| 8 | Accessories | Bike Racks | Hitch Rack - 4-Bike |
| 9 | Accessories | Bike Stands | All-Purpose Bike Stand |
| 10 | Accessories | Bottles and Cages | Mountain Bottle Cage |
| 11 | Accessories | Bottles and Cages | Road Bottle Cage |
| 12 | Accessories | Bottles and Cages | Water Bottle - 30 oz. |
| 13 | Accessories | Cleaners | Bike Wash - Dissolver |
| 14 | Accessories | Fenders | Fender Set - Mountain |
| 15 | Accessories | Helmets | Sport-100 Helmet- Black |
| 16 | Accessories | Helmets | Sport-100 Helmet- Blue |
| 17 | Accessories | Helmets | Sport-100 Helmet- Red |
| 18 | Accessories | Hydration Packs | Hydration Pack - 70 oz. |
| 19 | Accessories | Lights | Headlights - Dual-Beam |
| 20 | Accessories | Lights | Headlights - Weatherproof |
| 21 | Accessories | Lights | Taillights - Battery-Powered |
| 22 | Accessories | Locks | Cable Lock |
| 23 | Accessories | Panniers | Touring-Panniers- Large |
| 24 | Accessories | Pumps | Minipump |
| 25 | Accessories | Pumps | Mountain Pump |

**First & Last Order**

```sql
--Find the date of the first and last order
--how many years of sales are avaliable

SELECT
MIN(order_date) AS first_order_date,
MAX(order_date) AS last_order_date,
DATEDIFF(year,MIN(order_date),MAX(order_date)) AS Sales_Years
FROM fact_sales
```

No issues found

Results | Messages

| | first_order_date | last_order_date | Sales_Years |
|---|---|---|---|
| 1 | 2010-12-29 | 2014-01-28 | 4 |

**Youngest & Oldest Customer**

```sql
--Find the youngest and the oldest customer
SELECT
MIN(birthdate) AS oldest_customer,
DATEDIFF(year,MIN(birthdate),GETDATE()) AS Oldest_age,
MAX(birthdate) AS youngest_customer,
DATEDIFF(year,MAX(birthdate),GETDATE()) AS Youngest_age
FROM dim_customers

--4:MEASURES EXPLORATION
```

No issues found

Results | Messages

| oldest_customer | Oldest_age | youngest_customer | Youngest_age |
|---|---|---|---|
| 1916-02-10 | 110 | 1986-06-25 | 40 |

**Total Sales & Quantity**

```sql
--4:MEASURES EXPLORATION

-- Find the Total Sales
SELECT SUM(sales_amount) AS Total_sales FROM fact_sales

--Find how many items are sold
SELECT SUM(quantity) AS Total_quantity FROM fact_sales
```

100 % No issues found

Results Messages

| | Total_sales |
|---|---|
| 1 | 29356250 |

| | Total_quantity |
|---|---|
| 1 | 60423 |

**Avg Sales Price, Total Orders & Total Products**

```sql
--Find the average selling price
SELECT AVG(price) AS Avg_price FROM fact_sales

--Find the Total number of Orders
SELECT COUNT(DISTINCT order_number) AS Total_orders FROM fact_sales

--Find the total number of products
SELECT COUNT(product_name) AS Total_products FROM dim_products
SELECT COUNT(Distinct product_name) AS Total_products FROM dim_products
```

100 % No issues found

Results | Messages

| | Avg_price |
|---|---|
| 1 | 486 |

| | Total_orders |
|---|---|
| 1 | 27659 |

| | Total_products |
|---|---|
| 1 | 295 |

| | Total_products |
|---|---|
| 1 | 295 |

## Total Customer & Total Customer who give sales

```sql

--Find the total number of customer
SELECT COUNT(DISTINCT customer_key) AS Total_Customer FROM dim_customers

--Find the total number of customers that has placed an order
SELECT COUNT(DISTINCT customer_key) AS Total_Customer FROM fact_sales

--As we can clearly see that all  of our registered customers placed an order.

--Generate Report that shows all key metrics of the business. (Do All things in one query)
```

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Results | Messages

| | Total_Customer |
|---|---|
| 1 | 18484 |

| | Total_Customer |
|---|---|
| 1 | 18484 |

## An Overall Report

```sql
--Generate Report that shows all key metrics of the business  (Do All things in one query)
SELECT 'Total Sales' AS measure_name, SUM(sales_amount)AS measure_value FROM fact_sales
UNION ALL
SELECT 'Total Quantity', SUM(quantity) FROM fact_sales
UNION ALL
SELECT 'Average Price',AVG(price) FROM fact_sales
UNION ALL
SELECT 'Total No. Orders', COUNT(Distinct order_number) FROM fact_sales
UNION ALL
SELECT 'Total Products',COUNT(DISTINCT product_name) FROM dim_products
UNION ALL
SELECT 'Total No. of Customers' , COUNT(DISTINCT customer_key) FROM dim_customers
```

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Results Messages

| | measure_name | measure_value |
|---|---|---|
| 1 | Total Sales | 29356250 |
| 2 | Total Quantity | 60423 |
| 3 | Average Price | 486 |
| 4 | Total No. Orders | 27659 |
| 5 | Total Products | 295 |
| 6 | Total No. of Customers | 18484 |

**Countries**

```sql
-- 5:MAGNITUDE

--Combine multiple values to generate useful insights

--Find the Total Number of Customers by Countries
SELECT
country,
COUNT(customer_key) AS Total_Customers
FROM dim_customers
GROUP BY country
ORDER BY Total_Customers DESC
```

100 % No issues found

Results Messages

| | country | Total_Customers |
|---|---|---|
| 1 | United States | 7482 |
| 2 | Australia | 3591 |
| 3 | United Kingdom | 1913 |
| 4 | France | 1810 |
| 5 | Germany | 1780 |
| 6 | Canada | 1571 |
| 7 | n/a | 337 |

**Gender**

```sql
--Find total customers by gender
SELECT
gender,
COUNT(customer_key) AS Total_Customers
FROM dim_customers
GROUP BY gender
```

100 % No issues found

Results Messages

| | gender | Total_Customers |
|---|---|---|
| 1 | n/a | 15 |
| 2 | Male | 9341 |
| 3 | Female | 9128 |

**Total Revenue by each category**

```sql
--Find total products by category
SELECT
category,
COUNT(product_key) AS Total_Products
FROM dim_products
GROUP BY category
ORDER BY Total_Products DESC

--What is the average cost in each category
```

100 % No issues found

Results | Messages

| | category | Total_Products |
|---|---|---|
| 1 | Components | 127 |
| 2 | Bikes | 97 |
| 3 | Clothing | 35 |
| 4 | Accessories | 29 |
| 5 | NULL | 7 |

**Avg cost in each Category**

```sql
--What is the average cost in each category
SELECT
category,
AVG(cost) AS AVG_COST
FROM dim_products
GROUP BY category
ORDER BY AVG_COST DESC

--What is the total Revenue generated by each category
SELECT
```

100 % No issues found

Results | Messages

| | category | AVG_COST |
|---|---|---|
| 1 | Bikes | 949 |
| 2 | Components | 264 |
| 3 | NULL | 28 |
| 4 | Clothing | 24 |
| 5 | Accessories | 13 |

**Total Revenue**

```sql
--What is the total Revenue generated by each category
SELECT
p.category,
SUM(f.sales_amount) total_revenue
FROM fact_sales f
LEFT JOIN dim_products p
ON p.product_key=f.product_key
GROUP BY p.category
ORDER BY total_revenue DESC
```

100 % No issues found

Results Messages

| | category | total_revenue |
|---|---|---|
| 1 | Bikes | 28316272 |
| 2 | Accessories | 700262 |
| 3 | Clothing | 339716 |

**Top 5 Products**

```sql
--Which 5 products generate the highest revenue?
SELECT TOP 5
p.product_name,
SUM(f.sales_amount) total_revenue
FROM fact_sales f
LEFT JOIN dim_products p
ON p.product_key=f.product_key
GROUP BY p.product_name
ORDER BY total_revenue DESC

--Which are the 5 worst performance products in terms of sales
```

100 % No issues found

Results Messages

| | product_name | total_revenue |
|---|---|---|
| 1 | Mountain-200 Black- 46 | 1373454 |
| 2 | Mountain-200 Black- 42 | 1363128 |
| 3 | Mountain-200 Silver- 38 | 1339394 |
| 4 | Mountain-200 Silver- 46 | 1301029 |
| 5 | Mountain-200 Black- 38 | 1294854 |

**Least 5 products**

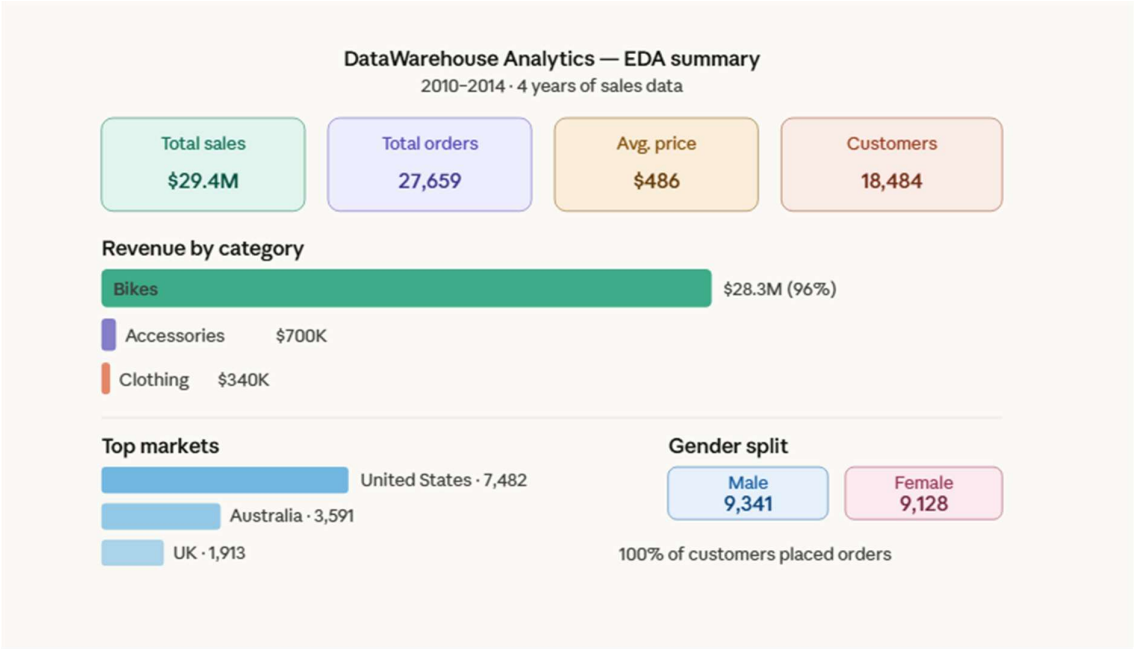
```sql
--Which are the 5 worst performance products in terms of sales
SELECT TOP 5
p.product_name,
SUM(f.sales_amount) total_revenue
FROM fact_sales f
LEFT JOIN dim_products p
ON p.product_key=f.product_key
GROUP BY p.product_name
ORDER BY total_revenue
```

100 % No issues found

Results | Messages

| | product_name | total_revenue |
|---|---|---|
| 1 | Racing Socks- L | 2430 |
| 2 | Racing Socks- M | 2682 |
| 3 | Patch Kit/8 Patches | 6382 |
| 4 | Bike Wash - Dissolver | 7272 |
| 5 | Touring Tire Tube | 7440 |

## Visual Summary



## Business Recommendations

- **Bikes dominance**: Bikes are 96% of revenue — that's a ticking time bomb. Start pushing accessories and clothing to every bike buyer before a slow season hits hard.
- **Mountain-200 star product**: Your best sellers are all Mountain-200s, so protect that line like it's your lifeline — bundle it, promote it, never let it go out of stock.
- **Dead weight products**: Racing Socks and Patch Kits are just collecting dust. Throw them in as freebies with bike purchases or cut them — they're not worth the shelf space.
- **Underserved European markets:** UK, France, and Germany have serious cycling cultures but low customer numbers. A small push there could open up a whole new revenue stream.
- **100% conversion rate:** Every registered customer already bought something — stop spending on new customer acquisition and focus on getting existing ones to come back again.
- **Gender split:** Your customers are almost exactly half male, half female. If your marketing looks like it's made for one gender, the data is telling you to fix that.